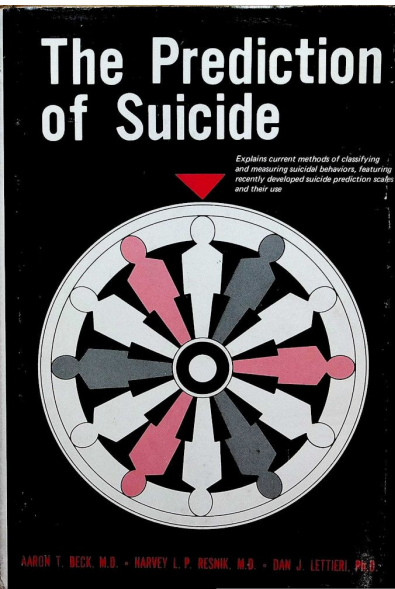


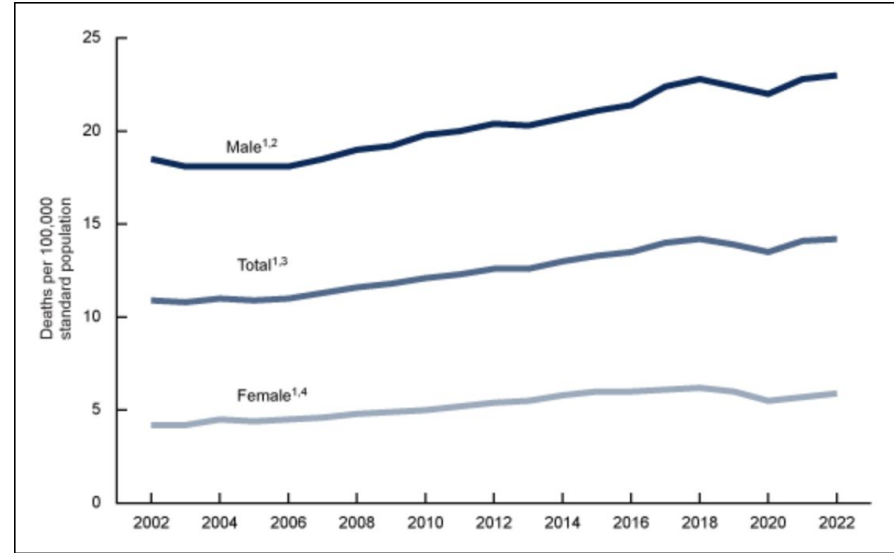
Suicide Text Detection Using NLP Model

By: Fardin Haiz, Anusha
Umashankar, Ritu Patel, Yeji Kim



Introduction

- Suicide is a very important problem that has seen increasing numbers
- With the increase in online presence of humans, there are signs that can be read.
- Inspired us to develop a tool that used different NLP models to classify suicidal text.



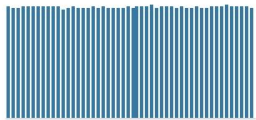
Source: <https://www.cdc.gov/nchs/products/databriefs/db509.htm>

Introduction Cont

- Developed a tool that used multiple models to classify text.
- Multiple models were used to classify data from the dataset into suicidal or not suicidal.
- The models :
 - TF-IDF + Logistic Regression
 - Long Short-Term Memory (LSTM)
 - BERT (Bidirectional Encoder Representations from Transformer)

Dataset

- Over 200k observations
- From Kaggle

#	text	class
	context extracted from user posts(Input)	target variable
 2 348k	232074 unique values	2 unique values
2	Ex Wife Threatening SuicideRecently I left my wife for good because she has cheated on me twice and ...	suicide
3	Am I weird I don't get affected by compliments if it's coming from someone I know irl but I feel rea...	non-suicide
4	Finally 2020 is almost over... So I can never hear "2020 has been a bad year"	non-suicide

Classical NLP Baseline Model Overview

Model: TF-IDF + Logistic Regression

Objective: Establish a strong and interpretable baseline for suicide ideation detection

Why Classical NLP?

- Fast and scalable for large text corpora
- Strong performance on sparse text features
- Produces interpretable term importance coefficients
- Serves as benchmark for LSTM and BERT comparisons

Key Idea: Represent text using TF-IDF features and classify using a linear decision boundary

Modeling Setup

Text Preprocessing

- Lowercasing, URL removal, stopwords filtering
- Lemmatization for vocabulary normalization

Feature Extraction: TF-IDF

- 100,000 max features
- N-grams: 1–3 to capture phrases like “want to die”

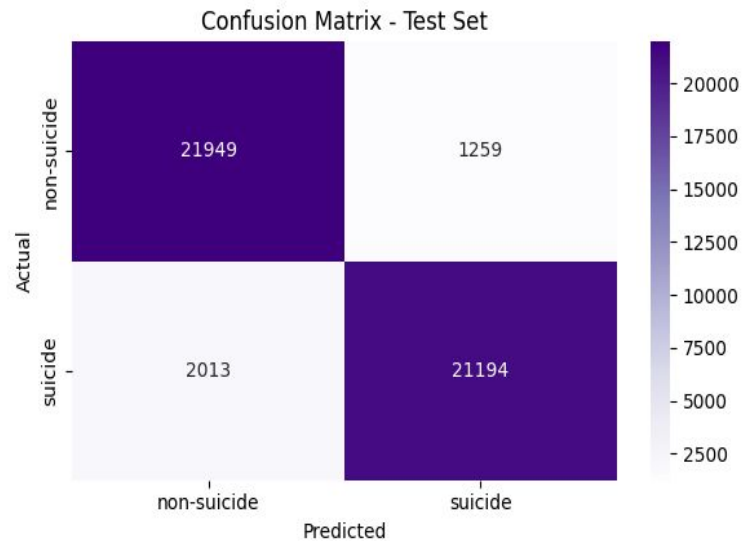
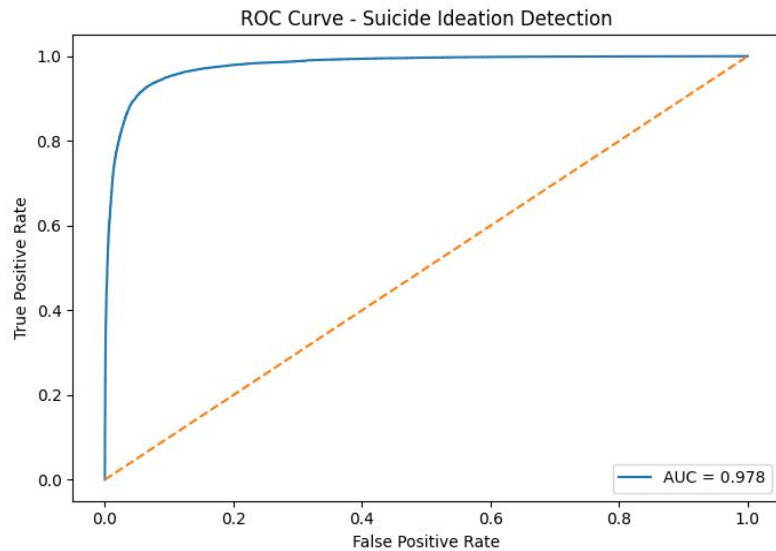
Classifier: Logistic Regression

- Sigmoid probability output
- Class balancing enabled to avoid bias

Training Configuration

- 80/10/10 train-validation-test split
- Hyperparameter tuning (GridSearchCV, macro-F1 scoring)

Evaluation Results



Dataset	Accuracy	Macro F1
Validation	0.94	0.94
Test	0.94	0.94

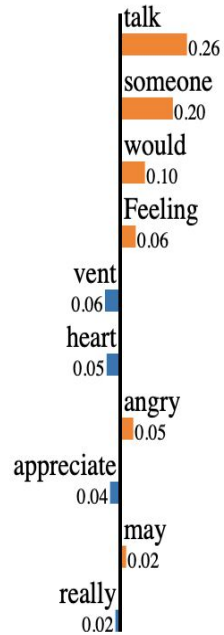
Explainability

Prediction probabilities



non-suicide

suicide



Text with highlighted words

Feeling really angry can someone talk? I would appreciate it if I may vent my heart..

Key Takeaways & Limitations

Strengths

- Fast and computationally efficient
- Interpretable predictions
- Strong baseline: 94% accuracy and macro-F1

Limitations

- Cannot fully capture long-range emotional context
- Struggles with sarcasm, metaphors, or indirect self-harm language
- No semantic understanding beyond bag-of-words patterns

Why LSTM?

RNN struggles with long-range dependencies

LSTM introduces: Memory cell, Feedback gate, Input gate, Output gate

Suitable for contextual risk cues.

Model Architecture

Embedding Layer

Bidirectional LSTM Encoder

- Captures forward + backward context

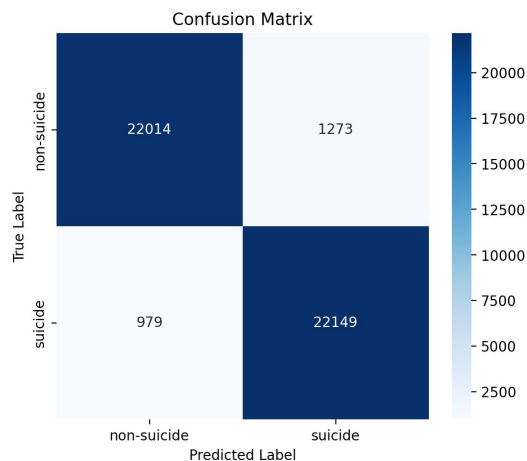
Attention Mechanism

- Learns importance weights for each token

Dense Softmax Layer

- Predicts 3 classes: Suicide / Depression / Other

Evaluation Results

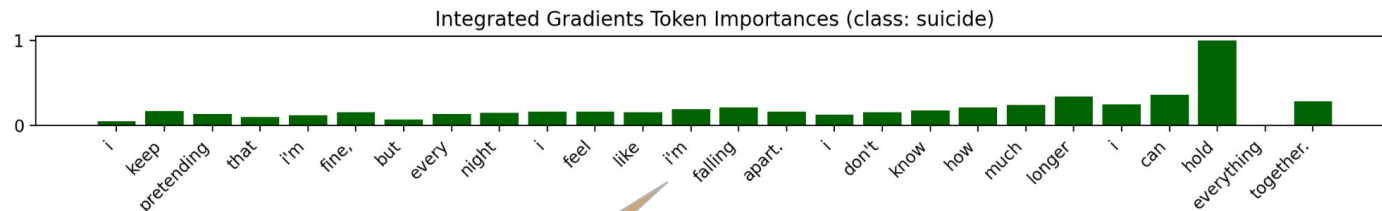


Metric	Score
Accuracy	0.9515
Macro Precision	0.9515
Macro Recall	0.9515
Macro F1-score	0.9515

Table 1: Final evaluation metrics on the validation set.

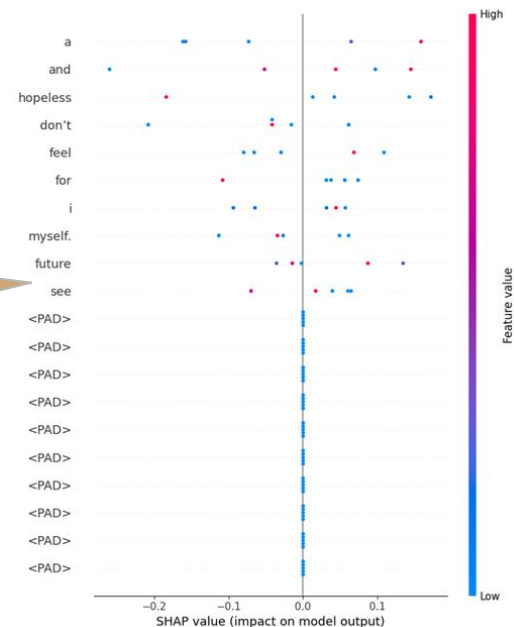
	precision	recall	f1-score	support
non-suicide	0.96	0.95	0.95	23287
suicide	0.95	0.96	0.95	23128
accuracy			0.95	46415
macro avg	0.95	0.95	0.95	46415
weighted avg	0.95	0.95	0.95	46415

Explainability



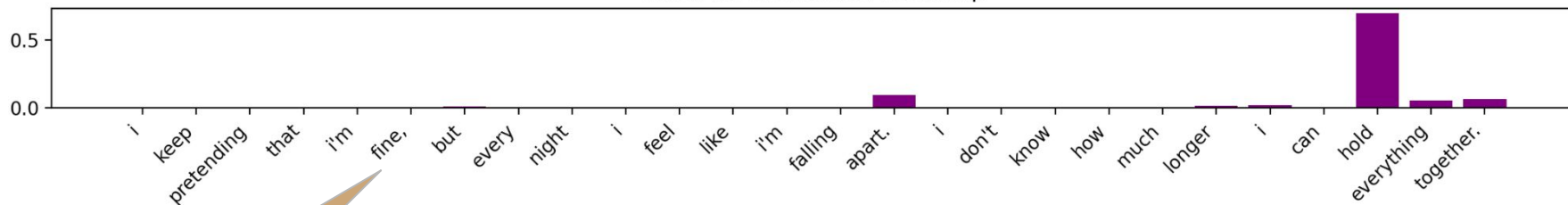
Integrated Gradients token importance for a suicidal input. Higher bars correspond to stronger contributions.

SHAP summary plot showing global feature importance across the dataset.



Attention Visualizations

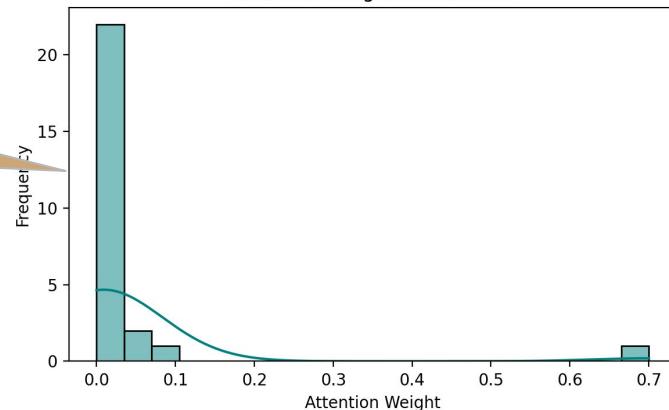
Token-level Attention Heatmap



Token-level attention heatmap for a suicidal post. Darker colors indicate higher attention weights.

Attention weight distribution showing sparsity and emphasis on emotionally charged words

Attention Weight Distribution



Summary of BiLSTM

- BiLSTM successfully captures contextual emotional cues
- Attention improves interpretability
- Explainability methods validate model's reasoning
- Model performs reliably across metrics
- Some misclassifications due to sarcasm, ambiguity, or short posts

BERT Model

- A transformer-based language model trained on large English text
- Reads sentences in **both directions**, so it understands context better than RNNs
- Well-suited for tasks where meaning depends on subtle wording or emotional tone

Why BERT

- Suicidal ideation often appears in **indirect or emotionally subtle ways**
- Social media contains slang, typos, and fragmented writing → BERT usually handles this better than classical NLP
- Provides a strong baseline for comparing error patterns
- Attention weights allow us to interpret what the model focuses on

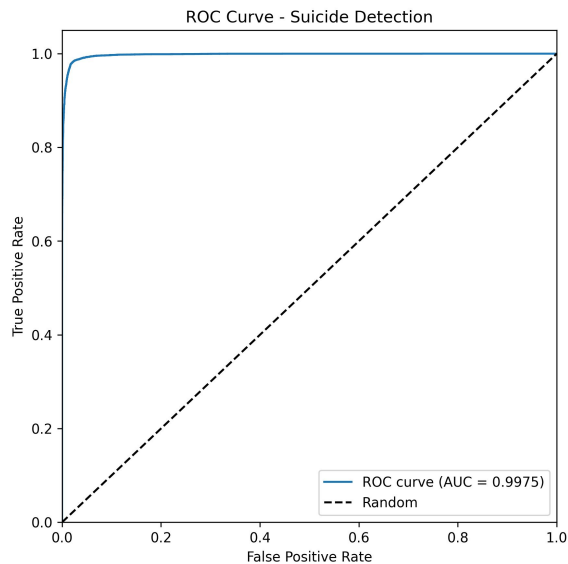
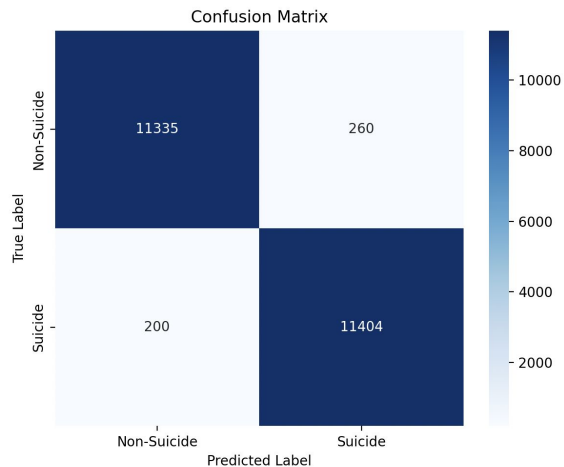
Model Setup

- Input format: [CLS] + tokenized post + [SEP]
- Maximum length: **320 tokens** (covers most posts)
- Extract the **[CLS] hidden state**, pass through:
 - Dropout (0.1)
 - Linear layer → output: Suicide / Non-suicide
- Training settings:
 - AdamW (lr=2e-5), batch size=16, 3 epochs
 - Early stopping to prevent overfitting

Evaluation Result

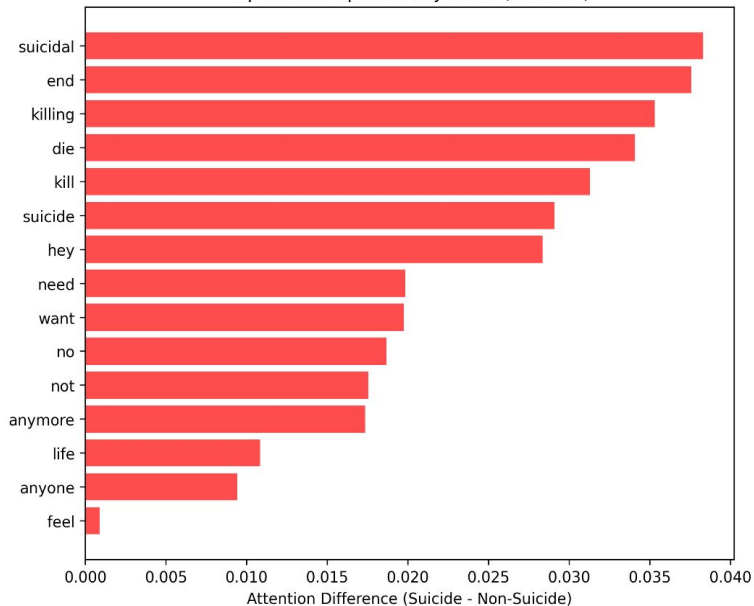
	precision	recall	f1-score	support
0	0.9827	0.9776	0.9801	11595
1	0.9777	0.9828	0.9802	11604
accuracy			0.9802	23199
macro avg	0.9802	0.9802	0.9802	23199
weighted avg	0.9802	0.9802	0.9802	23199

ROC_AUC: 0.9975

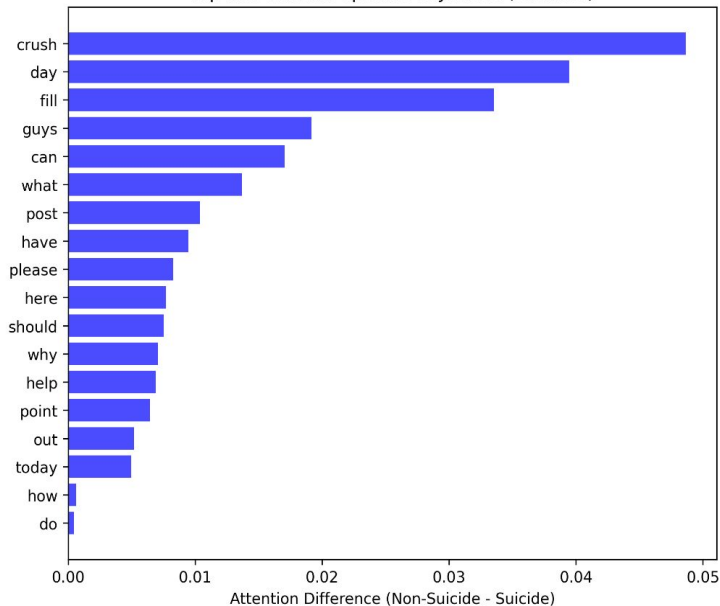


Explainability

Top Suicide-Specific Keywords (Cleaned)



Top Non-Suicide-Specific Keywords (Cleaned)



Key Takeaway & Limitation

- BERT relies heavily on clear contextual cues.
- When posts are short, vague, or slang-heavy, it becomes unreliable.

```
(base) MacBook-Air-3:bert_pipeline_pyproject yeji$ python scripts/99_infer.py \  
> --model_dir /Users/yeji/Desktop/bert_pipeline_pyproject/models/bert_base_uncased_single \  
> --text "im gonna kill myself"  
  
prob_suicide=0.0281, pred=0
```

```
(base) MacBook-Air-3:bert_pipeline_pyproject yeji$ python scripts/99_infer.py \  
> --model_dir /Users/yeji/Desktop/bert_pipeline_pyproject/models/bert_base_uncased_single \  
> --text "I feel completely hopeless and tired of living. im gonna to kill myself because I cannot take it anymore."  
  
prob_suicide=0.9989, pred=1
```

```
(base) MacBook-Air-3:bert_pipeline_pyproject yeji$ python scripts/99_infer.py \  
> --model_dir /Users/yeji/Desktop/bert_pipeline_pyproject/models/bert_base_uncased_single \  
> --text "I am going to kill myself."  
  
prob_suicide=0.9987, pred=1
```

Model Comparison

- The models all have good performance for different phrases
- The models have similar performance across phrases that are correctly formatted
- The BERT model struggles with incorrect grammar and non complete sentence structure

Conclusion

- We successfully developed a tool that classifies text through different models
- The user inputs the model and it outputs the classification based on each NLP model
- All of the models have good performance on various different phrases that can be inputted by the user.

Demo

Suicidal Text Detection — DATS 6312

Enter text below and click the button to see predictions from different models.

Input text:

Life is feeling very hopeless.

Run models

Results

	Model	Prediction	Confidence
0	Improved Baseline LR	suicide	0.9998
1	BERT	suicide	0.9984
2	LSTM + Attention	suicide	1.0000

<https://nlp-suicide-app-937335176181.us-east1.run.app/>